Math 1200 Lecture 16

Oct 29 2026

Topics: 16.4: Green's theorem, calculating area Name:_____
Collaborators:_____

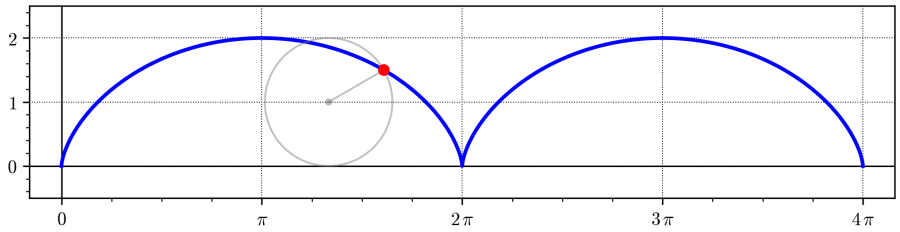
Group work 1 Draw the following regions with their positively oriented boundary

a) $D = \left\{ (x,y) \in \mathbb{R}^2 \;\middle|\; \begin{array}{l} x^2+y^2 \leq 1 \\ x \leq y \leq 2x \end{array} \right\}$ b) $D = \{(x,y) \in \mathbb{R}^2 \mid 1 \leq x^2+y^2 \leq 4\}$

Group work 2 Calculate the line integral $\int_C \langle e^{\cos(x)}, \sqrt{1-y^3}+3x \rangle \cdot d\vec{r}$ on the curve C where C is the triangle with vertices $(0,0),(0,1),(1,0)$ oriented counterclockwise.

Group work 3 Calculate the area of the quadrilateral Q with vertices $(3,0), (2,4), (-1,1), (1,0)$

Group work 4 A *cycloid* is the curve C traced by a point on a rolling circle. One loop is parametrized by $\vec{r}(t) = \langle t - \sin(t), 1 - \cos(t) \rangle$, $0 \leq t \leq 2\pi$. Write an integral for the



area under (one loop of) the cycloid, and evaluate if you have time.